Papers

Pharmaceuticalindustrysponsorshipandresearch outcomeandquality: systematicreview

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Abstrac

Objective To studies by with whether pharmaceu trials with other

Methods and searches in the Data were authors' consensus.

Results 30 drug research by have studies confidence None of reported poorer

Conclusio are made Explanatio inappropri investigate d and

Introdu

Clinicalres industryaffectshowdoctorspractisemedicine. increasingnumberofclinicaltrialsatallstagesina[1] favourable outcome may result in biases in design, out-come, and reporting of industry sponsored research.[7]

Arecentsystematicreviewoftheimpactof financialconflictsonbiomedicalresearchfoundthat studiesfinancedbyindustry,althoughasrigorousas other studies, always found outcomes favourable to the sponsoring company.[8] However, this review looked for paperspublishedonlyinEnglish, excludedreportsin lettersandabstracts,andlookedatstudiesfundedby other industries. We reviewed the relation between the sourceoffundingoftheresearchandthereported outcomesandinvestigatedwhetherqualityofthe methodsinstudiesfundedbypharmaceuticalcompa-nies differs from that in other studies.

Methods

Study selection

We included only studies that specifically stated that they analysed research sponsored by a pharmaceutical com-pany,comparedmethodologicalqualityoroutcomes with studies with other sources of funding, and reported theresultsinquantitativeterms.Outcomesofinterest were conclusions about differences in drug effectiveness, adverseeffects,costoutcomes,orpublicationstatus betweenindustryfundedtrialsandothertrials.Work published in any language was eligible for inclusion.

Somestudiesanalysedbothpharmacologicaland non-pharmacologicaltrialsandcombinedresearch fundedbydrugcompaniesandotherindustriesinto one group. In these cases, if most were non-pharmaceutical trials and were funded by other industries they were excluded.

Search strategy

An product'slifecyclearefundedbythepharmaceutical industry,[2,3] probably reflecting the fact that the pharma-ceutical industry now spends more on medical research

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We searched Medline from January 1966 to December 2002 using a combination of terms as both MESH subjectheadings(exploded)andkeywords("clinical trials,""conflictofinterest,""drugindustry,""financial support,""publicationbias"(subjectheadingonly), and "research design," and "research support.") We thandotheNationalIn States.[4]Mostpharmacoeconomicstudiesareeither donein-housebythedrugcompaniesorexternallyby consultants who are paid for by the company.[56]

Resultsthatareunfavourabletothesponsor—that is, trials that find a drug is less clinically effective or cost effective or less safe than other drugs used to treat the samecondition—canposeconsiderablefinancialrisks tocompanies. Pressuretoshowthatthedrugcausesa

searchedEmbasefromJanuary1980toDecember 2002 using a combination of terms as subject headings (exploded)andkeywords(“clinicaltrials”(subject headingonly),“drugindustry,”“ethics,”“financial management,”“methodology,”and“ethics.”Tofind more studies, we scanned the reference lists from each ofthearticlesandsearchedtheCochranemethod-ologyregister. Weplacedmessagesontwoemaildrug

discussiongroups,contactedcontentexperts,and searchedourpersonallibraries.Incaseswherethe reportedresultswereincomplete,wecontactedthe leadauthorandaskedforfurtherdetails.Asingle author(JL)didtheinitialselectionofstudiesandsent copiesofeachofthesestudiestotheotherthree authors for validation of the inclusion criteria.

excluded:
industry or funding

Data collection

From each study, we extracted the study design, type of researchassessedinthestudy,designofresearch assessedinthestudy,searchstrategyus edtolocate research,timeperiodcovered,drugord rugclass, disease,numberofindustryandnon-industryfunded articlesanalysedineachstudy,howind ustryfunding wasdefined,criteriausedtoassessmet hodological quality of the research, results with respect to method-ologicalqualityoroutcomeoftheresea rch,and primary purpose of study.

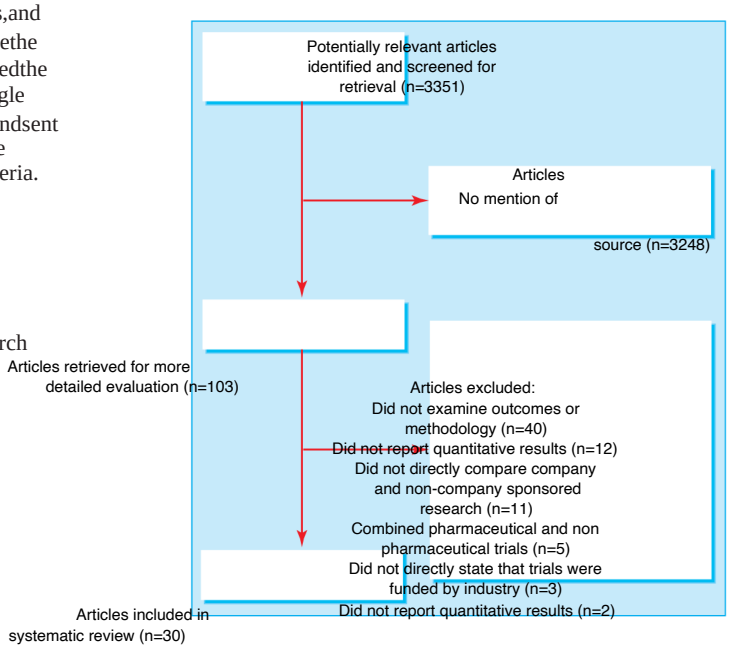


Fig1 QUOROMstatement

Weprovideacriticaldescriptionofeachi ncluded study,butdonotassessmethodologicalc uality(see table1).Sinceourincludedstudieshada rarietyof designs—thatis,cohortcollectionsoftrials,meta-analyses, and economic studies—and since we included letters and abstract with limited descriptions of meth-ods,wehadnovalidandreliablequalitya sessment instrumentavailableforassessingtheir methodologi-calquality.Wedidnotuseacomponenta proachto assesstheirqualitysincethisapproacha pliesto meta-analysis.[39] Wefoundnoothercasesofdouble counting, butassomeofthepapersdidnotprovidea fulllistofreferenceswecouldnotexcludethe possibility of further overlap.

Characteristics of included studies

Table1givesthecharacteristicsofthe30studies includedinthisanalysis.[12–41] Sixwerereviewsof pharmacoeconomicreports,[12232629343436] tworeviewed meta-analysesandsystematicreviews,[1825] andthe remaining 22 analysed groups of clinical trials.[13–1719–2224272830–333537–41] Elevenpapersmentioned that some trials were funded by industry but offered no further definition of industry funding.[17182224–26283436394] Intheother15papersthe definitionvariedfromastatementacknowledging industry funding in the[1232] article to a more comprehensive definition.[35]

We used the programStatsD[910] and considered P < 0.05 significant.

Results

Search results

Thecombined searche

Relationship between source of funding and outcome

Atotalof26ofthe30studiesreportedresultsonthe associationoftheoutcomeoftheresearchandthe source of funding:six examined the effects on publication,[17212431333] fivelookedattheoutcomeof pharmacoeconomicstudies,[1223263436] and16analysed theoutcomeofclinicaltrialsandmeta-analysesof clinical trials (table 2).

[13–1618–20222427303238–41]

funding. We read 103 articles in full (eight in languages other than English); we retained 30 articles for analysis. ReasonsforexclusionaredetailedintheQUOROM statement (fig 1).

The studies by Friedberg et al[23] and by Knox et al[29] analysedthesamesetof44trialsbutlookedat differentaspectsofthetrials:conclusionsaboutthe usefulness of products in one case,[23] and how the trials had been reported in the other.[29] Only one article was duplicatedinthetwostudiesreportedbytheg roup includingChard, Tallon, andDieppe[1318] (JChard, per-sonal communication, 2002). Seven of the nine articles in Kemmeren's[27] meta-analysis of third generation oral contraceptives were also included in Vandenbroucke's

Funding source and publication status

Researchfundedbydrugcompanieswaslesslikelyto bepublishedorpresentedthanresearchfundedby other sources (table 2).[12] Three studies looked at time topublication,[24 31 37] andtwoofthesefoundthat companysponsoredresearchtooklongertobe published than research with other sources of funding.[24 31] Researchfundedbydrugcompanieswas alsomorelikelytobepublishedintheproceedingsof[33] symposiums than non-industry sponsored research.

Funding source and economic outcomes

Pharmacoeconomicstudiessponsoredbythedrug industryweremorelikelytoreportresultsfavouring

Table1 Generalcharacteristicsofstudies

Study (firstauthor)	Typeofpublication	Researchassessed	Designofresearch	Searchstrategy usedtofind research	Timecovered	Numberof separatetrials fundedbyindustry	Numberof separatetrialsnot fundedbyindustry	Howwasindustry fundingdefined
Azimietal[12]	Journalarticle, systematicreview	Pharmacoeconomic studies	Costeffectiveness	AbridgedIndex Medicus	1January1990to 30June1996	10	34	Explicit acknowledgmentof supportfrom privateindustryin article
Chardetal[13]	Journalarticle, systematicreview	Clinicaltrials	Prospective experimental, retrospectiveor prospectivecoh studies,general disease reviews/ specific intervention reviews,studies reportingpooled data	BIDSInstitutefor Scientific Information, CochraneLibrary, Embase,Medline	1950toMarch 1998	114with pharmaceutical sponsorshipout 128with commercial sponsorship	802	Sourceoffunding declaredin publishedarticle
Choetal[14]	Journalarticle, systematicreview	Clinicaltrials	Experimental (randomisedan non-observational	Computergenerate listofrandom numbersfrom625 symposiums, symposiums' articlesmatchedto articlefromparent journal	Notstated	Forqualityscore 50;foroutcome 40	Forqualityscore 77;foroutcome: 112	Article acknowledgeddr companyprovidin fundingordrugsor anyauthors employedbydrug company
Cliffordetal[15]	Journalarticle, review	Clinicaltrials	Randomised controlled	Manualsearchf randomisedtria 5generalmedic journals (convenience sampleof100 trials;20/journal)	January1999to October2000	44bycompany only;22by companyandat least1notfor profitsource	28	Acknowledgedgrant support;drug companyemployee listedasauthor; drugsuppliedby company;other typesofcompany support
Davidson[16]	Journalarticle, review	Clinicaltrials	Concurrentor crossovercontr group,beforea after	Searchof5major generalmedicine journals	1984	37	70	Acknowledgment statedsupportfrom company(excluding supplyofdrugor placebo)orany authorsworkedfor company
Dickersin etal[17]	Journalarticle, cohortstudy	Clinicaltrials	Observational, clinicaltrial,oth er	Manualsearch InstitutionalReview Boardlogsfor studiesreceiving approvalatone university	Studiesapprovedin 1980orbefore 1980andstill ongoing	42	472	Notstated
Dieppeetal[18]	Journalletter, review	Meta-analyses	Meta-analysis	Searchfor meta- high- impact journals	1993to1997	18	117	Notstated
Djulbegovic etal*[19]	Abstract,system review	Clinicaltrials	Randomised controlled	Fromlistoftrials maintainedby AmericanSoci Hematology	1980to2000	23	7	Sourceoffunding obtainedfrom paperor investigator; associationwith manufacturerif drugsuppliedor grantsupport
Djulbegovic etal*[20]	Journalarticle	Clinicaltrials	Randomised controlled	Cochranesear strategy	1996to1998	35	101	Sourceoffunding obtainedfrom paperor investigator; associationwith manufacturerif drugsuppliedor grantsupport
Easterbrook etal[21]	Journalarticle, retrospectivecoh study	Clinicaltrials	Observational, experimental (clinicaland non-clinical)	Manualsearch ethicscommittee recordsat1 university	1January1984to 31December1987	108	177	Directcontactwit investigatoror co-investigator
Freemantle etal[22]	Journalarticle, systematicreview	Clinicaltrials	Randomised controlled	Embase,Medline, reviewedrefere lists,contacte d authors	1982to1997	Notstated	Notstated(tota 105trials)	Notstated
Freidberg etal[23]	Journalarticle, systematicrevie w	Pharmacoeconomic studies	Costminimisation, costidentification, costeffectiveness	HealthStar,Medline	1988to1998	20	24	Statementinarticle orbycontacting investigatorifno statement
Ioannidis[24]	Journalarticle, retrospectiveco study	Clinicaltrials	Randomised	Trialsconducte underauspices twoAIDStrial groups	1986to1996	11	98	Notstated

Continuedonnextpage

Table 1 General characteristics of studies — *continued from previous page*

Study (first author)	Type of publication	Research assessed	Design of research	Search strategy used to find research	Time covered	Number of separate trials funded by indu	Number of separate trials not funded by industry	How was industry funding defined
Jadad et al[25]	Journal article, systematic review	Systematic reviews, meta-analyses	Systematic review, meta-analysis	CINAHL, Cochrane Library, Embase, HealthStar, Medline, personal collections, reference lists	1988 to 1998	6	44	Not stated
Kamal-Bahl et al[26]	Journal article, systematic review	Pharmacoeconomic studies	Cost effectiveness, cost benefit, incremental cost analysis	HealthStar, Medline, contacted experts, reference lists, Cochrane Database of Systematic Reviews, National Health Service Economic Evaluation Database, National Institute for Clinical Excellence, Canadian Coordinating Office of Helath Technology Assessment	1 January 1990 to 31 August 2001	4	8	Not stated
Kemmeren et al[27]	Journal article, meta-analysis	Clinical trials	Cohort, case-control	Medline, contacted experts, reference lists	October 1995 to December 2000	4	5	Industry funding explicitly stated in acknowledgement
Kjaergard et al[28]	Journal article, retrospective coh study	Clinical trials	Randomised controlled and quasi-controlled	Handsearch of *Hepatology*, Medline	January 1981 to August 1998	Not stated	Not stated (tota 235)	Not stated
Knox et al†[29]	Journal article, systematic review	Pharmacoeconomic studies	Cost minimisation, cost identification, cost effectiveness	HealthStar, Medline	1988 to 1998	20	24	Statement in article or by contacting investigator if no statement
Koeppe et al[30]	Journal letter, comment on meta-analysis	Clinical trials	Not stated	Studies were reported in previously published meta-analysis	Not stated	8	5	Authors employed by manufacturer, received grants, held stocks or options, in-house research of methodological or statistical analysis provided by manufacturer
Liebeskind et al*[31]	Abstract, system review	Clinical trials	Randomised controlled	Medline search international tri registries	1957 to 1997	Not stated	Not stated (tota 127)	Not stated but review discriminated between corporate sponsorship and corporate provision of drug
Mandelkern[32]	Journal letter, review	Clinical trials	Not stated	Manual review one year of reg issues of *Journal of Clinical Psychiatry*	1997	16	16	Stated at front of article
Massie et al§[33]	Journal letter, review	Clinical trials	Not stated	Articles published over 3 year period in 8 major journals	1980 to 1982	59	23	From article and by direct contact with authors
Neumann et al[34]	Journal article, systematic review	Pharmacoeconomic studies	Cost effectiveness	Medline, HealthStar, CancerLit	1975 to 1997	cost effectiveness 77; quality 32§	cost effectiveness 538; quality 118§	Not stated
Rochon et al*[35]	Journal article, systematic review	Clinical trials	Randomised controlled	Medline search of 3 journals	January 1990 to November 1992	142 by company only; 25 both company and government/ foundation	16 government/ foundation; 59 no source	Acknowledged grant support; drug company employee listed as author; drugs supplied by company; other types of company support
Sacristan et al*[36]	Journal letter, review	Pharmacoecon studies	Cost effectiveness	Medline search in 6 general medical journals; hand search of *Pharmacoeconomics*	Medline: 1988 to 1994; *Pharmacoeconomic* 1993 to 1994	General medical journals 6; *Pharmacoeconomic* 18	General medical journals 63; *Pharmacoecomonic* 6	Not stated
Sternet al*[37]	Journal article, retrospective cohort study	clinical trials	Randomised controlled, concurrent and historical controlled, uncontrolled	Search ethics committee submissions at 1 hospital	September 1979 t December 1988	Not stated	Not stated	Stated in submission to ethics committee

Continued on next page

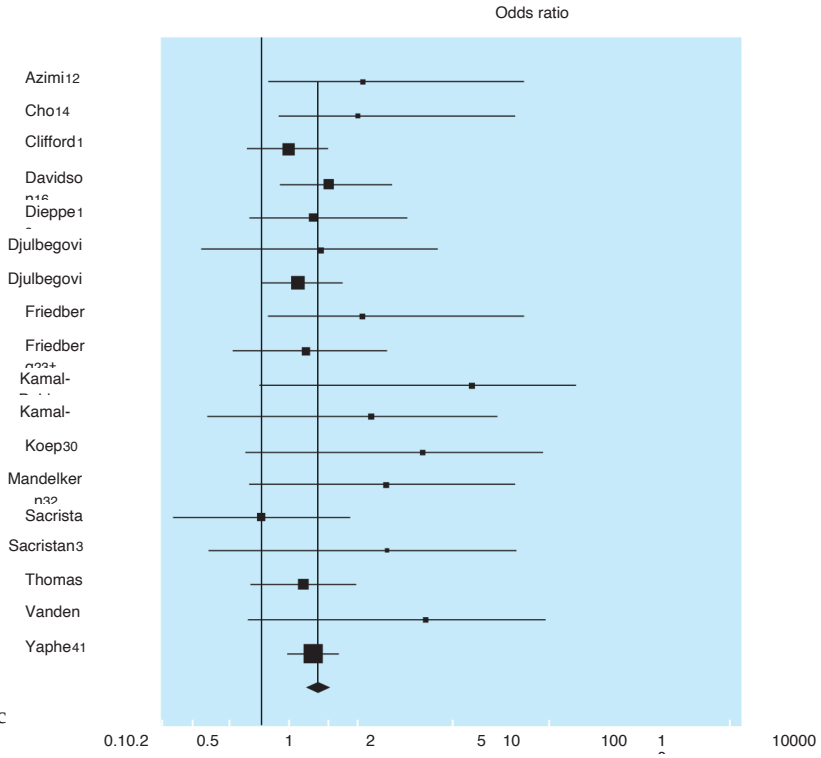
Table1 Generalcharacteristicsofstudies— *continuedfromprevious page*

Study (firstauthor)	Typeofpublication	Researchassesse	Designofresearch	Searchstrategy usedtofind research	Timecovered	Numberof separatetrials fundedbyindu	Numberof separatetrialsnot fundedbyindustry	Howwasindustry fundingdefined
Thomasetal[38]	Journalletter, systematicreview	Clinicaltrials	Originalresearch	Medline	1966to2001	48(30bycompa A;18bycompan B)	22	Fundedby company;author wascompany employee; correspondence addressused companyaddress ; statisticalanalysis donebycompany
Vandenbroucke etal[39]	Journalletter, review	Clinicaltrials	Notstated	Notstated	Notstated	4	9	Notstated
Wahlbeck etal[40]	Journalletter, commenton systematicreview	Clinicaltrials	Notstated	Cochranesear strategy	Notstated	16	13	Notstated
Yapheetal[41]	Journalarticle, systematicreview	Clinicaltrials	Randomised controlled	Handsearchof majormedical journals	8October1992to 1October1994	209	96	Fundedby company;provision ofstudymaterials bycompany;author wascompany employee;statistical analysisdoneby company

*Additionaldatafromauthor.
†AdditionaldatafromFreidbergetal. [23]
‡AdditionaldatafromMassieetal. [42]
§Forcosteffectiveness:numberofratios(studiescontainedmorethanoneratio;70ratiosfrompharmaceuticalcompanyfundedstudiesand7fromdevicecompanyfundedstudies);for quality:numberofstudies.

thesponsor'sproductthanstudieswithothersources offundinginallfivearticlesthatexaminedthis question.[1,22,32,6,34,38] Inthreecases,however,thebiasin favourofindustryfundedresearchdependedonthe particularquestionbeingposed[23,26] oronwherethe pharmacoeconomic analyses were published.[36]

analysesofclinicaltrials. Thesummaryoddsratiowas 4.05 (95% confidence interval 2.98 to 5.51).



Funding source and outcomes of clinical trials and meta-analyses

Sixteenstudiesinvestigatedtherelationshipbetween fundingsourceandtheoutcomesofclinicaltrialsand meta-analyses.Ofthese,13foundthatclinicaltrials andmeta-analysessponsoredbydrugcompanies favouredtheproductproducedbythefunder. Statistical significance for this finding was reported in eight of the 13 studies,[13,14,16,19,20,38,40,41] and in another two there wasatrendtowardsstatisticalsignificance[18,22] These studies covered a wide range of diseases, such as arthritisoftheknee,[13,18] multiplemyeloma,[20] various psychiatricproblems,[32,40] Alzheimer'sdisease,[30] and venousthromboembolism,[39] andawiderangeof drugs,suchastacrine,[30] clozapine,[40] thirdgeneration oral[39] contraceptives,[19] erythro antidepressants,[22] andtopicalglucocorticosteroids.[38] Onestudy thatfoundnodifferencelookedattheoutcomeof trialsoftreatmentforHIVandassociatedcomplicationsandinthiscasethetrialsweremonitoredbythe National Institutes of Health.[24] In one meta-analysis thirdgenerationoralcontraceptives,[27] theriskof venousthromboembolismfornon-industryfunded researchwashigherthanthatforindustrysponsored trials, although the increased risk for thromboembolic diseasewassignificantinbothcases.Anotherstudy found no difference in outcomes in research published in five leading medical journals.

Figure 2 shows the individual odds ratios and sum-mary odds ratio for 18 different comparisons (15 studies) of the outcomes of ind non-industryfundedstudies—sevenfrompharmaco-economic analyses and 11 from clinical trials or meta-

Fig2 Sourceoffundingandoutcomeinpharmacoeconomicanalyses,clinicaltrials,and meta-analysesofclinicaltrialsofdrugtreatments;forreferencesseebmj.com(*Favourable qualitativeresults;†Overstatementofquantitativeresults;‡Reportingpossibilityofcost effectivenessorcostsavingsofprophylaxisinentirehighriskinfantpopulationeitherinpoint estimatesorsensitivityanalysis;§Reportingcosteffectivenessorcostsavingsineitherentire highriskpopulationsorspecificinfantsubgroupscomparedacrossstudies;¶Analyses reportedingeneralmedicaljournals;**Analysesreportedin *Pharmacoeconomics*)

Table2 Relationbetweensourceoffundingandoutcomeofresearch

Author	Outcomequestionassessedbyresearch	Results
Fundingsourceandpublicationstatus		
Dickersinetal[17]	Factorsassociatedwithpublicationoffindingsfromc	Publicationrateatoneoftwocentreswasconsiderablyhig fundedbyNationalInstitutesofHealththanforstudiesfundedbydrug industry(90.5% *v*65.0%),butnoindicationthattherewasanydifference inthetendencytopublishsignificantresults
Easterbrooketal[21]	Areclinicaltrialswithstatisticallysignificantresultsm publishedandwhatisthemagnitudeofthisbias	Drugcompanysponsoredtrialsofanytypearesignificantlylessli

	bepublishedorpresentedcomparedtounfundedstudies(OR =0.36;95% CI0.16to0.81);drugcompanysponsoredclinicaltrialsaresignificantly lesslikelytobepublishedorpresentedcomparedtounfundedstudies (0.17;0.05to0.53)	
Ioannidis[24]	Istimetocompletionandtimetopublicationofclini statisticalsignificanceofresults;descriptionofna	Trialswheredatamanagedbyindustryofshorterdurationtha federallysponsored(P<0.01)butnodifferenceintimetoappeara peerreviewedliterature(P =0.33)†
Liebeskindetal[31]	Evidenceofpublicationbiasinrandomisedcontrolle	Mediantimebetweenenrolmentandpublicationofind withpositiveresults3.5years *v*4.7yearsfornegativestudies;time differenceforstudieswithanytypeoffunding3.5 *v*4.4yearsforindustry fundedstudies
Massieetal[33]	Whethersourceoffundingofclinicaltrialsonanaina process	48%ofdrugcompanyfundedtrialspublishedinsymposiumsco with26%ofstudieswithothertypesoffundingorwheresource fundingnotstated
Sternetal[37]	Extenttowhichpublicationisinfluencedbystudyo	Pharmaceuticalindustryfundingnotastatisticallysign timetopublication
Fundingsourceandeconomicoutcomes		
Azimietal[12]	Howoftendocosteffectivenessanalysesencoura additionalexpenditures	Industryfundedstudiesmorelikelytosupportastrategyrequir additionalexpenditurescomparedwiththosewithoutsuchfundi 15/34,P=0.01);amongthe39articlesthatsupportedadditional expenditurethemediancosteffectivenessratioofthe9studiesfundedby industrywassignificantlyhigherthanfortheremainder(P =0.02)
Freidbergetal[23]	Relationbetweendrugcompanysponsorshipandec oncologydrugs	Drugcompanysponsoredstudiesmorelikelytoreportfavoura qualitativeconclusions(19/20 *v*15/24,P=0.04);overstatemen quantitativeresultsnotsignificantlydifferent(6/20 *v*3/24,P=0.
Kamal-Bahletal[26]	Relationbetweendrugcompanysponsorshipandec drugsforrespiratorysyncytialvirusprophylaxis	Industryfundedstudiesmorelikelytoreportpossibilityofcost

			effectivenessorcostsavingsofprophylaxisinentirehighriskinfant populationeitherinpointestimatesorsensitivityanalysis(4/4 v0/8, P=0.002);whenlikelihoodofreportingcosteffectivenessorcostsavingsin eitherentirehighriskpopulationsorspecificinfantsubgroupscompared acrossstudies,nodifferencebetweenindustryfundedandnon-industry fundedstudies(4/4 v3/8,P=0.08)
Neumannetal[34]		andpercentcostsavingreported	Relationbetweendrugcompanysponsorshipandc Forincrementalcosteffectivenessratio:industryfundedstudies

	non-industryfundedstudies$13,000(P =0.003);forpercentcostsaving: industryfundedstudies21% *v*non-industryfundedstudies9% (P=0.002)‡		
Sacristanetal* [36]	Relationbetweendrugcompanysponsorshipandre effectivenessstudies	Generalmedicaljournals:3/6cost-fundinghadpositiveresults *v*31/63withnofundingorother funding; *Pharmacoeconomics*:18/18costeffectivenessstudieswithindustry fundinghadpositiveresults *v*4/6withnofundingorothersourceof funding	
Fundingsourceandoutcomesofclinicaltrialsandmeta-analyses			
Chardetal [13]	Resultsofclinicaltrialsforosteoarthritisoftheknee	Projectscommerciallyfundedmorelikelytosupport non-commerciallyornon-specifiedfundingsourcestudies(P =0.024)*	
Choetal [14]	Relationbetweendrugcompanysponsorshipofc outcome	Proportionoftrialswithfavourableoutcomehigherindrugco sponsoredresearchthanintrialswithoutcompanysponsors 89/112,P<0.01)	
Cliffordetal [15]	Relationbetweenfundingsourceandtrialoutcome	Studiesfavouringnewproduct:industryonlyfundi industryandnon-industryfunding16/22;notforprofitfunding15/28 (P=0.461)	
Davidson [16]	Relationbetweendrugcompanysponsorshipofc outcome	Studiessupportedbycompaniessignificantlymorelikelytosupp therapiesthantrialswithothersourcesoffundingorwherefundin stated(33/37 *v*43/70,P=0.002)	
Dieppeetal [18]	Relationbetweendrugcompanysponsorshipand meta-analysesoftreatmentofosteoarthritisoftheknee	Trendformeta-significantbeneficialeffectcomparedwithmeta-offundingorunknownsourcesoffunding(16/18 *v*81/117,P=0.084)	
Djulbegovicetal [19]	Relationbetweendrugcompanysponsorshipofclinic erythropoietinandstudyoutcome	Drugcompanyfundedtrialsshowedpositiveresultsin21/23stud (P<0.0001)comparedwith5/7studiesfundedpartiallyor publicresources(P =0.096)	
Djulbegovicetal [20]	Relationbetweendrugcompanysponsorshipofclinic treatmentofmultiplemyelomaandstudyoutcome	Drugcompanyfundedtrialsshowedpositiveresultsin26/35st (P=0.004)comparedwith54/101studiesfundedbynon-organisations(P =0.608)	
Freemantleetal [22]	Investigatepotentiallyconfoundingfactorswhichma thatassessedtherelativeefficacyofantidepressants	Mostimportantstructuralpredictorofoutcomewastrial sponsorship—(coefficient0.097;95%CI −0.03to0.23)	
Ioannidis [24]	Istimetocompletionandtimetopublicationofclinicaltr statisticalsignificanceofresults;descriptionofcours	Nosignificantcorrelationbetweenpresenceofstatisticalsigni accrual,andwhetherdatamanagedbyindustryornot(allcorre coefficientsP<0.2)§	
Kemmerenetal [27]	Relationbetweendrugcompanysponsorshipandou comparingeffectsofsecondandthirdgenerationoral riskofvenousthromboembolism	Oddsratio1.3(95%CI1.0to1.7)forstudiesdirectlyfundedby pharmaceuticalindustry *v*2.3(1.7to3.2)inotherstudies	

Continuedonnextpage

Table2 Relationbetweensourceoffundingandoutcomeofresearch— *continuedfromprevious page*

Author	Outcomequestionassessedbyresearch	Results
Koeppetal[30]	Relationbetweendrugcompanysponsorshipofclinicaltrialson andstudyoutcome	0/5studieswithoutcorporatesupportfoundclinicalbenefit;6/7 withcorporatesupportfoundbenefit(1studycouldnotbelocat
Mandelkern[32]	Relationbetweendrugcompanysponsorshipofclinicaltrialsinp andstudyoutcome	16/16drugcompanyfundedstudiesreportedfavourableoutcom non-industryfundedstudiesreportedfavourableoutcome
Thomasetal[38]	Relationbetweendrugcompanyfundingstudiescomparingtop glucocorticosteroidsandwhetherstudyfavoursproductmade sponsoringcompany¶	Producttestedfavouredintrial:37/48incompanyfundedtrials *v*1 trialswithothersourcesoffunding(P<0.001)
Vandenbrouckeetal[39]	Relationbetweendrugcompanysponsorshipandoutc comparingeffectsofsecondandthirdgenerationoralcontraceptiveson riskofvenousthromboembolism	1/9studieswithoutindustryfundingfoundnohigherriskofve

thromboembolism(relativerisk1.5-4.0,summaryrelativerisk2.4);4/4 industryfundedstudiesfoundnohigherrisk(relativerisk0.8-1.5, summaryrelativerisk1.1)

Wahlbecketal[40]	Relationbetweendrugcompanysponsorshipofclinandstudyoutcome	Oddsofrelapsingsignificantlyinfavourofclozapineindrugco sponsoredtrials(oddsratio0.5;95%CI0.3to0.7)comparedwit non-sponsoredtrials(0.4;0.1to1.4);bothdrugcompanysponsoredand non-sponsoredtrialssuggestedclozapinemediatesclinicallyimportant improvementcomparedwitholderdrugsbutindustrysponsoredtrials weremorepositive(0.4;0.2to0.7 v0.3;0.1to0.7);industrysponsored trialsreportedsignificantlyfewerearlydropoutscomparedwitholderdrugs (0.5;0.4to0.7)butnon-sponsoredtrialsdidnot(0.6;0.3to1.2)
Yapheetal[41]	Relationbetweensourcesofsupportofresearcha ofrandomisedcontrolleddrugtrials	Positiveoutcomes:181/209trialswithindustryfunding v62/96 non-industryfunding(oddsratio3.45,95%CI1.90to6.62)

*Resultsforall128commerciallysponsoredtrials(114fundedbypharmaceuticalindustry).
†56completedtrials(of98wheredatawasnotmanagedbypharmaceuticalindustry)includedinanalysis.
‡70trialsfundedbypharmaceuticalcompaniesand7fundedbydevicecompanies.
§90trialswherefullpatientaccrualcompleted(of98wheredatawasnotmanagedbypharmaceuticalindustry)includedinanalysis¶Out comeofstudiesassessedbyindependentevaluators.

Relationship between source of research (table 3).

Atotalof13studiesexaminedtherelationship between source of funding and the methodological quality of the research[12131518192224262730323436 38–40] studies. Our results are also supported by Rochon and coworkers[43] (we found that drug com-

theofthe13reportedthatindustryfundedstudieshad poorermethodologicalquality.Oftheninethat provided statistical analyses, four[14–16192025282931–35] None

excludedthispaperbecauseallofthetrialswere sponsoredbydrugcompaniesandwere,therefore, notcomparablewithtrialslackingcompanyfunding.) They found that trials supported by manufacturers of non-steroidalanti-inflammatoryagentsalmostalways reported that the sponsor's drug was as or more effec-

pany sponsored research had better quality tive and less toxic [14283133] scores.

Nine of the studies on clinical trials used well established methods of assessing quality.[14151920242829313 5] The single study that reported on the methods of pharmacoec analyses used commonly[34]

One study evaluated the appropriateness of the comparators in clinical trials and found that a greater proportion of industry sponsored studies compared innovative treatment to either placebo or no therapy than did studies sponsored by public resources (60% v 21%; P < 0.001).[20]

Discussion

Research sponsored by the drug industry was more likely to produce results favouring the product made by the company sponsoring the research than studies funded by other sources. The results apply across a wide range of diseases states, drugs, and drug classes, over at least two decades and regardless of the type of

Explanations

At least four possible explanations exist for favourable results seen in industry sponsored research. Firstly, pharmaceutical companies may selectively fund trials on drugs that they consider to be superior to the competition. Data collected so far, however, indicate that[44] researchers cannot predict results of trials in advance.

Secondly, positive results could be the consequence of poor quality research conducted by industry. For example, low quality trials exaggerate the benefits of treatment by an average of 34%.[45,46] We found that the research methods of trials sponsored by drug companies is at least as good as that of non-industry funded research and in many cases better. This does

research being assessed—pharmacoeconomic studies, clinical trials, or meta-analyses of clinical trials. The totality of the evidence reported in our meta-analysis of a subset of homogeneous studies suggests that there is some kind of systematic bias to the outcome of pub-lished research funded by the pharmaceutical industry.

Our results confirm and extend those reported by

not guarantee the absence of bias in studies sponsored bytheindustrysinceoutcomecouldbeinfluencedby factorsleftoutofqualityscores,suchasthequestion[747] asked or the conduct or reporting of the study.

Thirdly,selectinganappropriatecomparatorisa key issue in planning a clinical trial.[72044] In the study by Rochonetal,inmostcasesinwhichthedosesofthe studyandcomparatordrugswerenotequivalent,the drug given at the higher dose was that of the support-ingmanufacturer.[43] Astheauthorssaw,higherdoses maybiastheresultsinfavourofeffectivenessofthe manufacturer's product. Safer also reports that in trials of psychiatric drugs the comparator drug is often given in doses outside substantial dose

Bekelmanetal.[8] Theyidentifiedonlyfivestudiesthat comparedoutcomesinresearchfundedbyphar- bythesponsorin[1416202341] maceuticalcompaniesandothersources, and our study adds another 16 researchfunded bythecompany marketingfluco na-

Table3 Relationbetweensourceoffundingandmethodologicalqualityofresearch

Study	Criteriausedtoassessmethodologicalqualit	Results
Choetal[14]	22itemvalidatedscoringsystem	Studydesignindrugcompanysponsoredclinicaltrialsb researchwherenostatedsponsorship(P =0.04)
Cliffordetal[15]	5itemvalidatedscoringsystem(Jadad)pluscomp onJadadscaleandadequacyofconcealment)app roach	Nodifferencebyfundingsourceforadequacyofallocationcon (P=0.377);nodifferencebyfundingsourceforoverall/ Jadadscale(P =0.143)
Davidson[16]	Samplesize,blinding	Foralltrialshigherrateofblindingforoneswithindustrysponsor (67.5%v41.8%,P=0.01);fortrialsinvestigatingmedicationsnodifference inblinding(P =0.46);forsamplesizenodifferencebetweenclinicaltrials supportedbydrugcompaniesandthosewithothersourcesoffundingor wherefundingnotstated
Djulbegovicetal[19]	5itemvalidatedscoringsystem(Jadad)	Nodifferenceinqualityscoresbetweenrando solelybyindustry(mean3.3(SD1.4);median:3.5)andtrialssupportedby publicsources(mean2(SD0.96);median:2)(P =0.308)
Djulbegovicetal[20]	5itemvalidatedscoringsystem(Jadad)	Randomisedcontrolledtrialsfundedsolelyor towardshigherqualityscores(mean2.94(SD1.3);median:3)thantrials supportedbygovernmentorothernon-profitorganisations(mean2.4(SD 0.8);median:2)(P =0.06)
Jadadetal[25]	7pointvalidatedscoringsystem(GuyattandOxma	6/6industryfundedsystematicreviewsandmeta- flawsversus34/44non-industryfundedreviews
Kjaergardetal[28]	5pointvalidatedscaleincluding:concealmentofallo allocationsequence,doubleblinding,dropouts/	Clinicaltrialsfundedbyeitherdrugordeviceindustryhadhigherq thantrialswithnoexternalfunding(P<0.001);qualityofpubliclyfu trialssameastrialsfundedbydrugordeviceindustry(P =0.68)
Knoxetal[29]	9itemscaledevelopedforthisstudy,includingclini generalisabledatasources,statisticaltestsofsignificanceperformedon appropriateoutcomes,statementregardingperspective,descriptionofcost s ofthemainincludedresources,descriptionoftimehorizon,descriptionof	Drugcompanysponsoredpharmacoeconomicanalyseslesslik sourceoftotalcostsdifferences,disc ussionoflimitations,comparisons withotherpublishedstudies

formallyreportonstudygeneralisability,butweremorelikelytoprovide informationonthekeycomponentsofthemethodssectionthanwere non-profitsponsoredanalyses

Liebeskindetal[31]	100pointscaleaddressing5aspectsoftrialdesignan randomisation,outcome,inclusion/ therapeuticregimen,statisticalanalysis	Clinicaltrialswithcorporatesupporthadbetterqualitythantrials non-	
Mandelkern[32]	Presenceorabsenceofplacebocontrol	3/16non-industryfundedtrials	5/16industryfundedclinicaltrialshadplaceb
Massieetal[33]	Notstated	Higherproportionofindustryfundedclinicaltrialswerea controlledanddesignedthanweretrialswithothersourcesoffunding (71%*v*33%,P<0.01)	
Neumannetal[34]	Adherencetorecommendedprotocolsforcosteffec (adequatedescriptionofalternatives,studyperspectiveclearlystated, discountedbothcostsandQALYsifneeded,incrementalanalyses		Nodifferencebetweenindustryandnon- performedcorrectly)plusqualityasj udgedbyreaders(scaleof1to7)

	measure:adequatedescriptionofalternativesP =0.30;studyperspective clearlystatedP =0.98;discountedbothcostsandQALYsP =0.65; incrementalanalysesperformedcorrectlyP =0.73;qualityasjudgedby readersP =0.49		
Rochonetal[35]		ModifiedversionofChalmersscoreincluding14ite	Nodifferenceinqualityscorebetweenindustryonlyfundedclinic
	and/orregimen,randomisation,blinding,patientsblinded,observers blindedtotreatmentandresults,previousestimateofnumbers,testing compliance,resultsofrandomisationonpretreatmentvariablesand		inclusioninanalysis,majorendpoint s,post-betaestimate,confidence

andthosefundedbygovernmentorfoundations(mean36.9%(SD17.6%) *v*37.1%(17.8%),P =0.271)

limits,statisticalanalyses,withdrawalsafterrandomisation,sideeffects discussion

QALY=qualityadjustedlifeyear.

zolecompareditwithoralamphotericinB,adrug known to be poorly absorbed, thereby creating a bias in favouroffluconazole.[49]Wedidnotconsiderwhois finally responsible for the selection of the comparator—investigators, regulatory agencies, or sponsors.

Finally,ourresultssuggestthatpublicationbias may explain our finding of bias in favour of outcomes ofresearchfundedbyindustry.Althoughresearch sponsoredbyindustrywaslesslikelytobepublished thanresearchwithothersourcesoffunding,thetwo studieswiththisfindingdidnotspecificallyexamine whethernon-publicationappliedjusttoresearchwith non-significantoutcomes.[17 21]Inthepastfewyears,

Limitations

Our study has several limitations,primarily the difficultyinlocatingresearchexaminingtheeffectsof companysponsorship.OurMedlineandEmbase searchesfoundonly13ofthepapersthat we manufacturershaveattemptedtoprevents tudies

least equal to those in studies funded by other sources, theabsenceo

included,[16–18202327–303238404] andtheremaining17
came from a search of the author's personal files, sug-
gestionsbyoutsideexperts,orscanningofreference
whichareunfa lists.[12–1519212224–263133–3739] The inability to critically
published in several high profile cases.[50–52]
Massie and colleagues raise another possible
source of publication bias.[33] They showed that research
fundedbyindustryappearsmoreofteninsymposi-
ums.Studiesinsymposiumsareknowntolackpeer

evaluate the mtehodology in the abstracts and journal lettersisanotherpossiblesourceofbias,andthe 3040
conclusionsoftwooftheletterstheatweincluded 5455
have been criticised.

Methodsinstudiessponsoredbyindustrywereat

review and to favour the sponsor's 1453product. leastasgoodasinstudieswithothersourc

Althoughthemethodsofindustryfundedtrialsareat funding.Conclusionsaboutoverallqua

tycanbe

Whatisalreadyknownonthistopic

When a pharmaceutical company funds research into drugs, studies are likely to produce results favourable to the sponsoring company

Whatthisstudyadds

Research funded by drug companies was more likely to have outcomes that favour the sponsor's product than research funded by other sources

This cannot be explained by the reported quality of the methods in research sponsored by industry

The result may be due to inappropriate comparators or to publication bias

influencedbytheinstrumentused,andsomeofthe scales may have missed important criteria.

Researchsponsoredbythepharmaceuticalindustryisfacinganumberofchallenges.Questionshave beenraisedaboutthemismatchbetweentheresearch agendasofthepharmaceuticalindustryandconsumers of research.[9] Meta-analysts are confronted with the problems of duplicate publication of data from

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company funded trials and the withholding of data.[495758]

Leadingmedicaljournalsrecentlydecidedto establishmorerigorouscriteriafortheacceptanceof researchsponsoredbyindustry;thisisastepinthe rightdirectiontowardsincreasingthecredibilityof studiespaidforbydrugcompanies.[56] Therevised CONSORTstatementshouldalsohelpimprovethe quality of clinical research.[5960] In addition, authors and editorsshouldconsiderincludingastatementconcerningpriorbeliefsoftheinvestigatorsaboutthe uncertainty of the treatments that are reported. Finally, allclinicaltrialsshouldberegisteredprospectivelyas

the only way to prevent publication bias.[61] The proposal to do so which was put forward in 1986[62] has been periodically renewed,[63–65] but to this date has not been implemented.

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